

WebUp Industrial Engineering - Documentation

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1. Introduction

Welcome to the WebUp Industrial Engineering documentation. This project is a modern, accessible, and SEO-optimized static website showcasing metal fabrication and industrial engineering projects.

1.1. Project Overview

WebUp is a professional HTML5/CSS3 website designed for industrial engineering companies specializing in metal fabrication, CNC machining, and manufacturing services.

Key Features

- Responsive design for all devices

- Accessibility compliant (WCAG standards)
- SEO optimized with meta tags and sitemap
- Privacy compliant (robots.txt, noindex tags)
- Modern CSS with variable-based theming
- Interactive image gallery with paper shadow effects
- Fixed footer with clean navigation

1.2. Technology Stack

Technology	Description
HTML5	Semantic markup with ARIA labels
CSS3	Modern styling with CSS variables
Git	Version control on branch v0.1.0
AsciiDoc	Documentation format

2. Getting Started

2.1. Prerequisites

- Web browser (Chrome, Firefox, Safari, Edge)
- Text editor (VS Code recommended)
- Basic knowledge of HTML/CSS

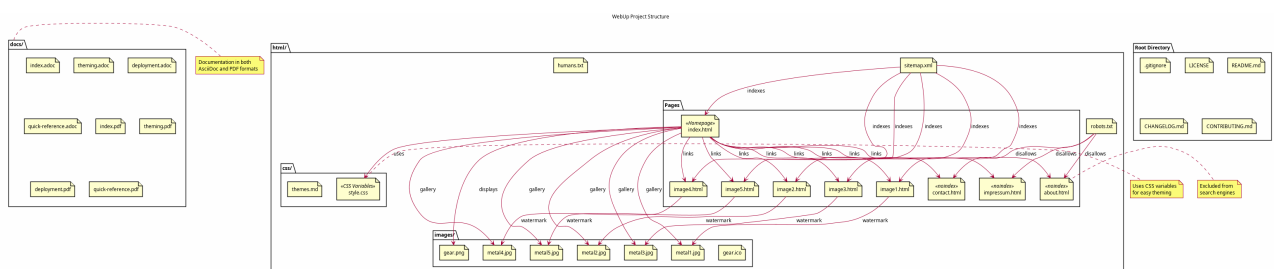
2.2. Installation

1. Clone the repository:

```
cd webup
```

- ## 2. Switch to the v0.1.0 branch:

```
git checkout v0.1.0
```



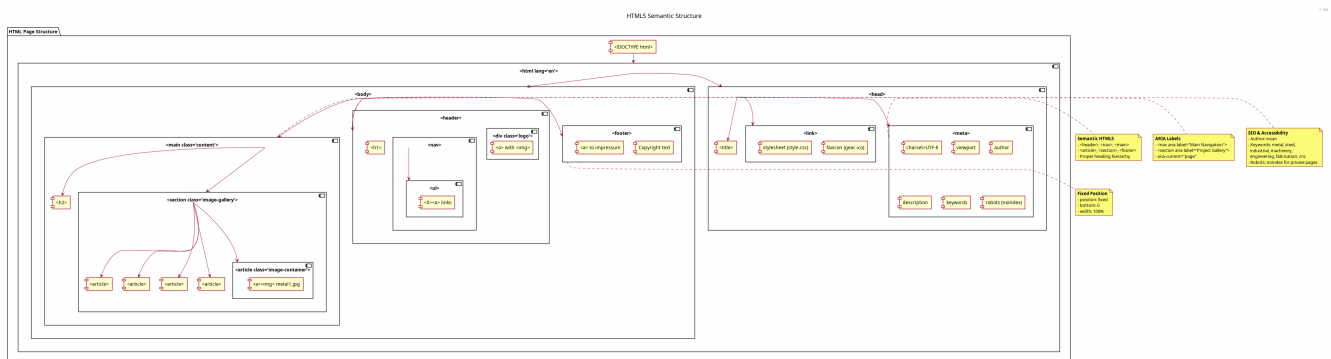
```
cd html
python -m http.server 8000
```

1. Navigate to <http://localhost:8000>

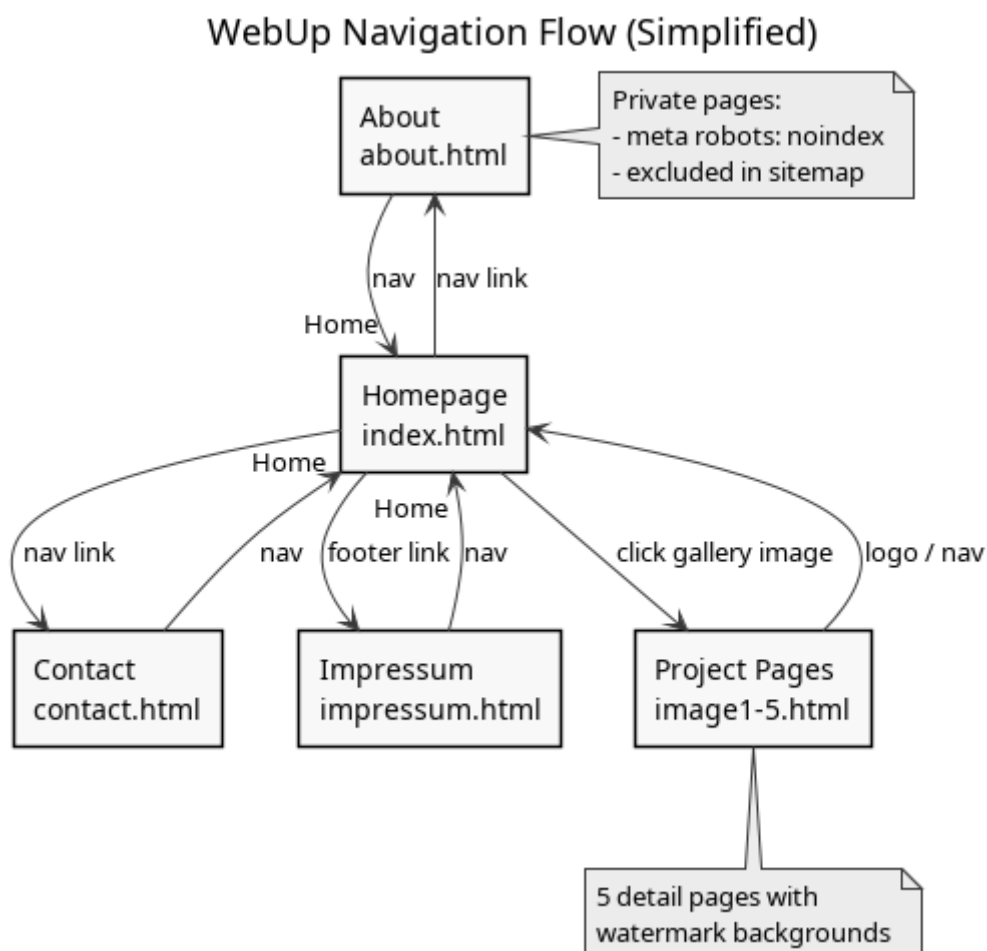
2.3. Quick Start

Open any HTML file in the `html/` directory directly in your browser. No build process required!

3. Project Structure

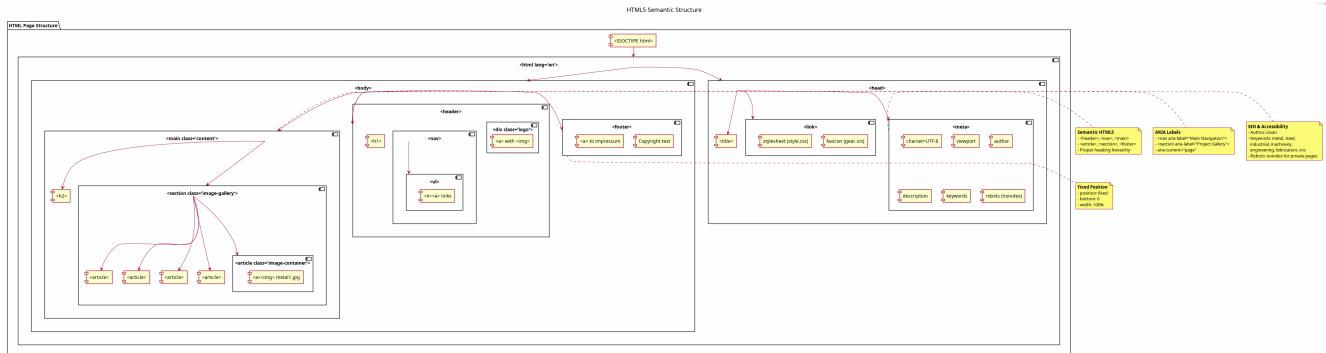


3.1. Navigation Flow



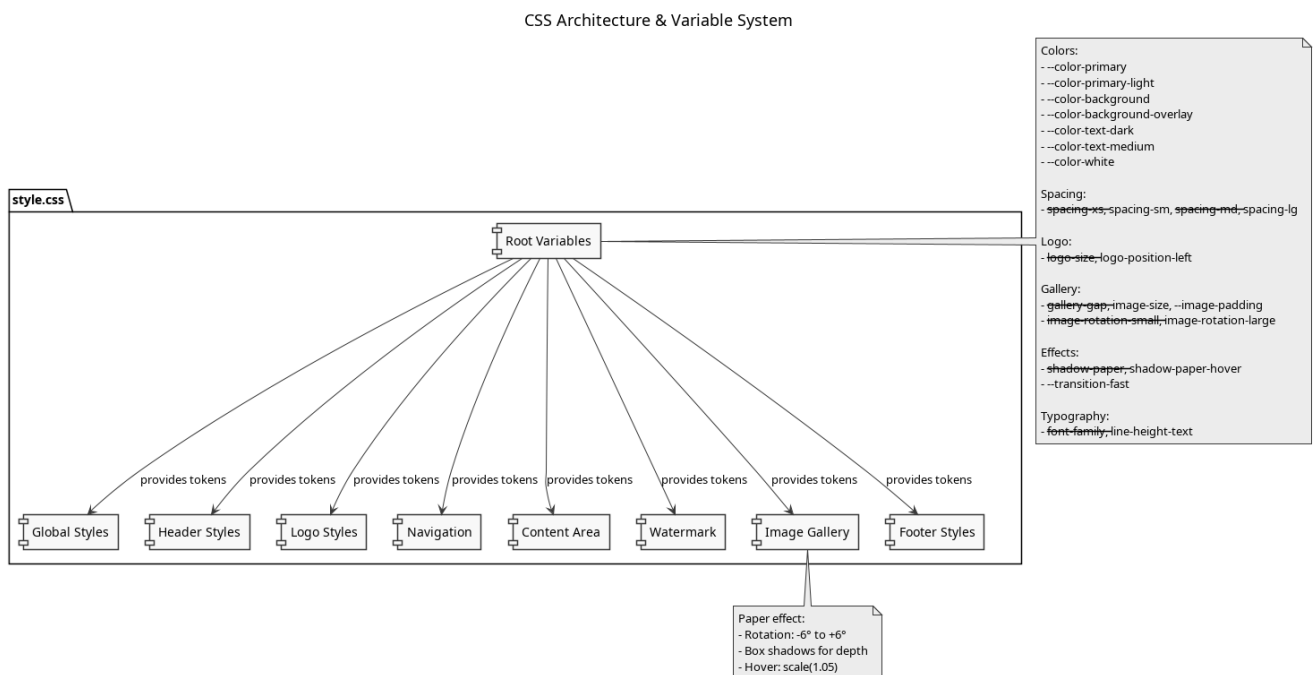
The diagram shows top-level files (licensing, meta, documentation) and the split between `docs/` and `html/` application assets.

4.2. Semantic HTML Skeleton



Every page shares the same accessible structure: `<header>` (logo + nav), `<main>` (content/articles), `<footer>` (legal + navigation anchor).

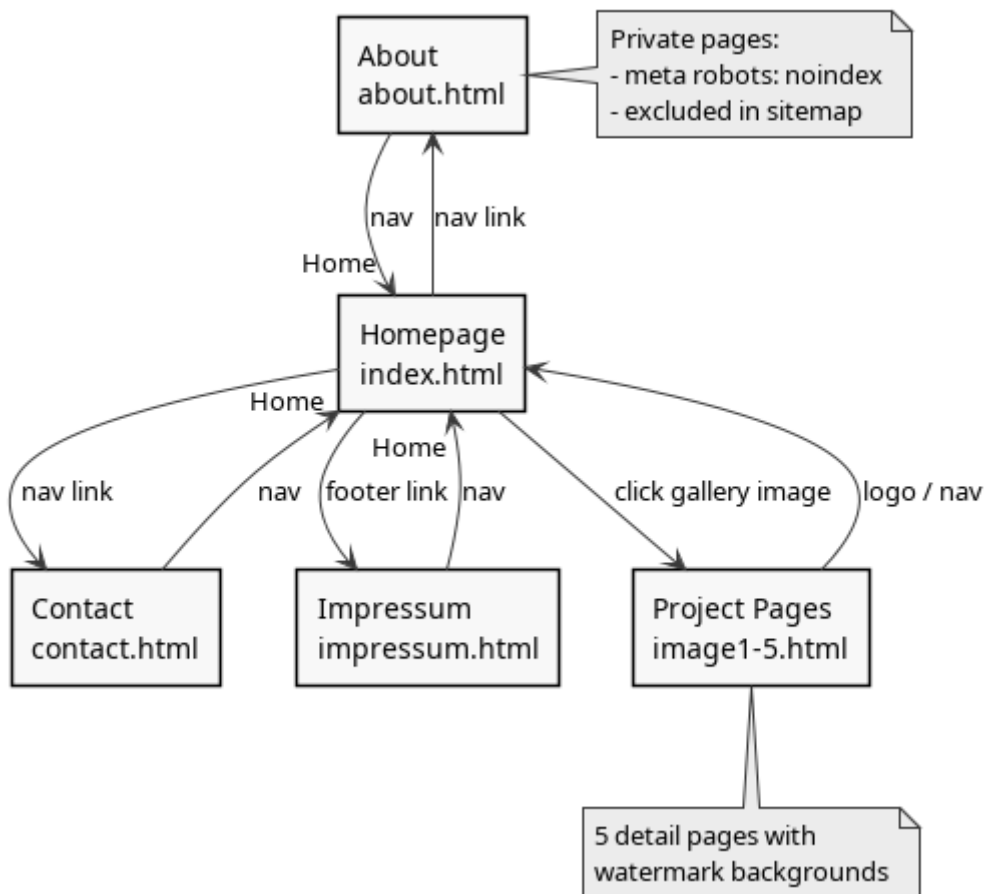
4.3. CSS Variable & Layer Architecture



Central design tokens in `:root` flow into each style section enabling rapid theming and consistent spacing, color, and effect reuse.

4.4. Navigation Flow

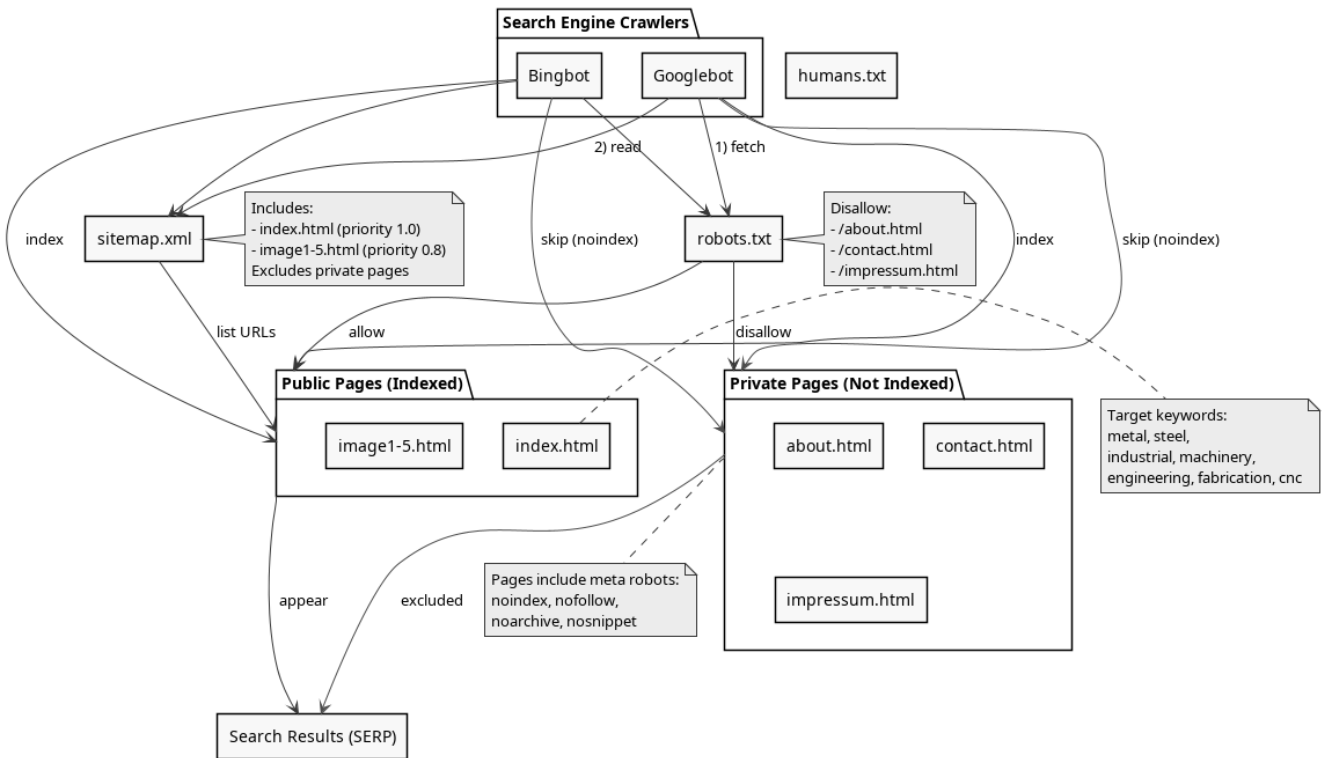
WebUp Navigation Flow (Simplified)



Public gallery drives traffic into project detail pages; private (noindex) informational pages and impressum remain excluded from search but reachable via global nav/footer.

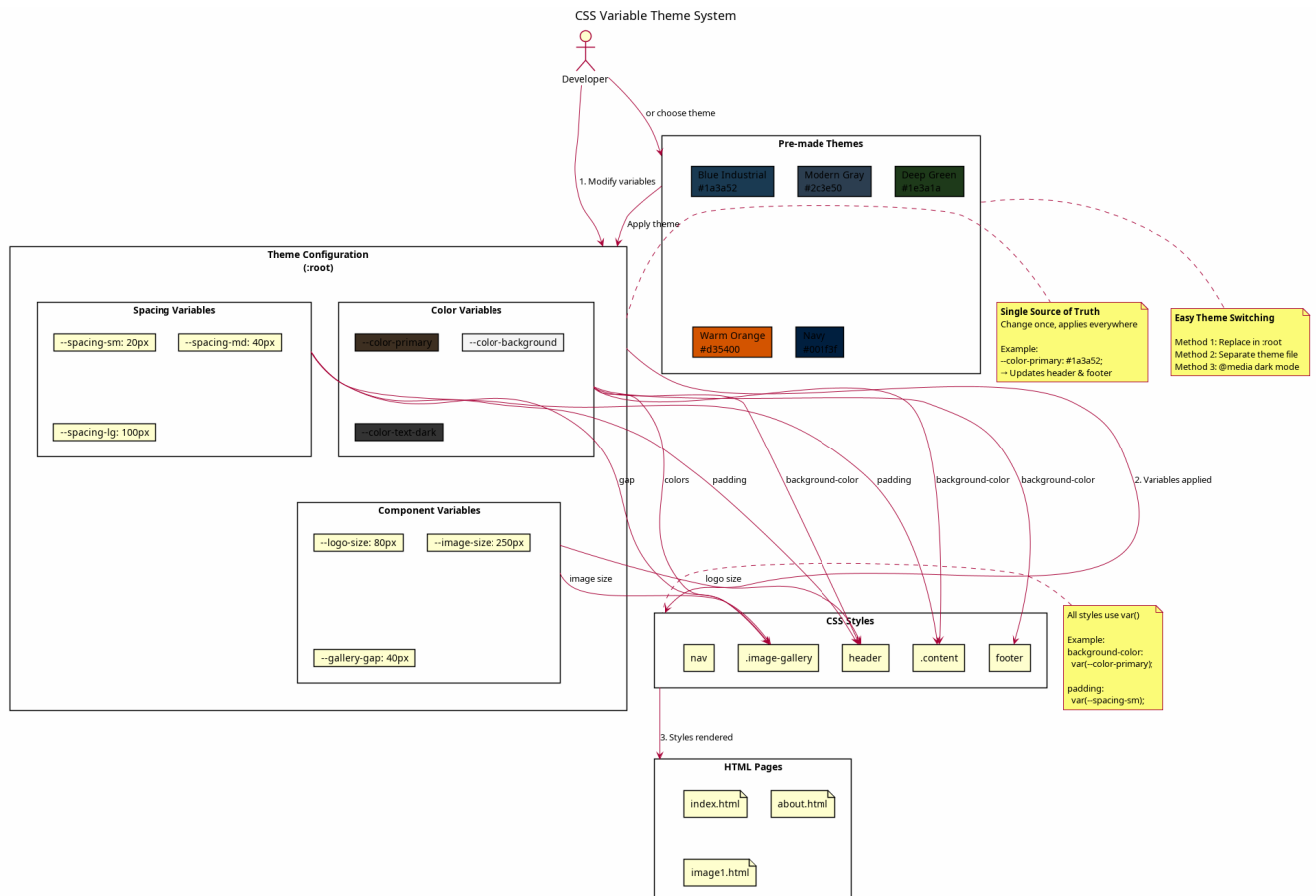
4.5. SEO & Privacy Architecture

SEO & Privacy Architecture



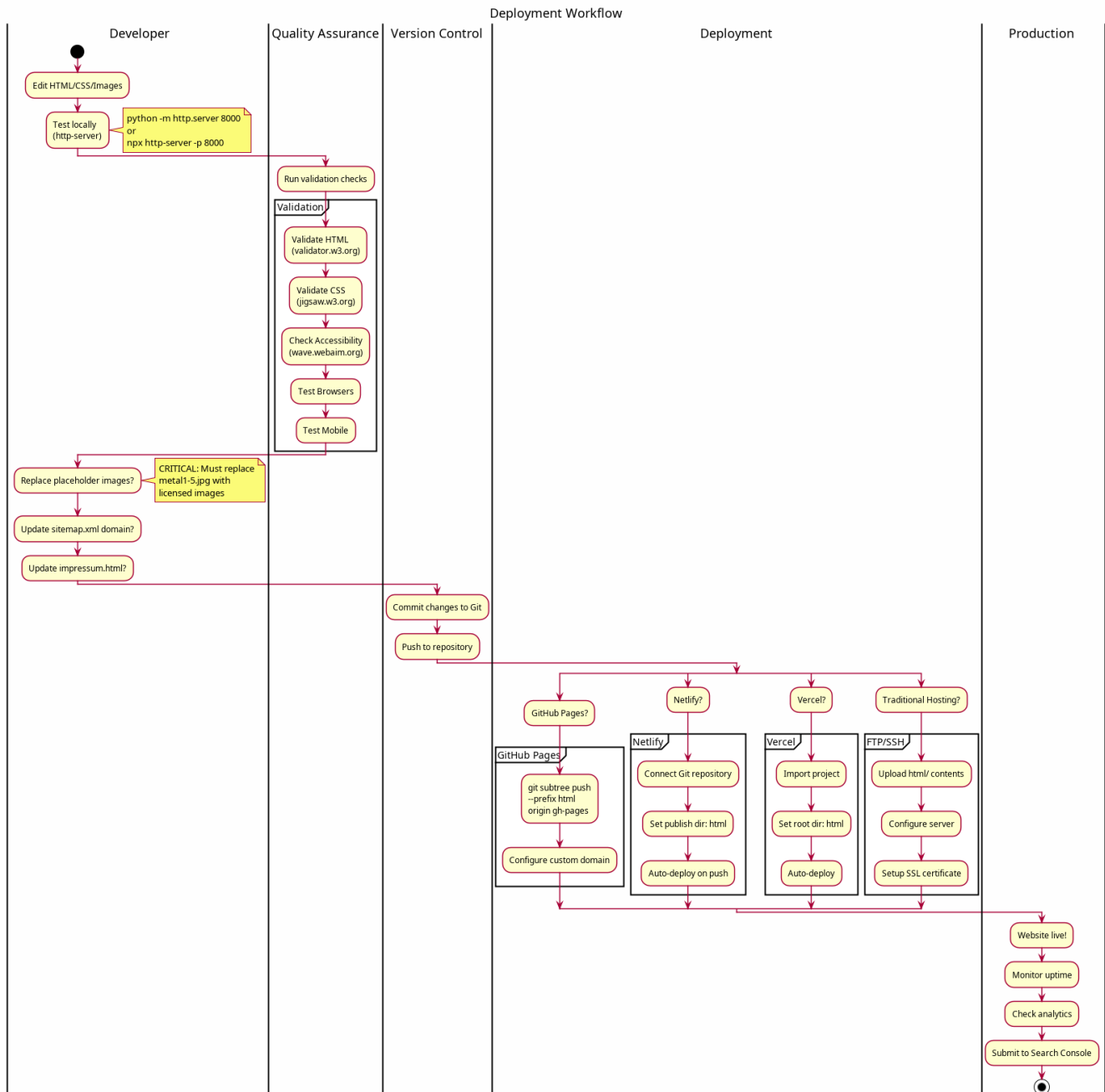
robots.txt and meta robots directives gate crawlers; **sitemap.xml** lists only public pages ensuring controlled search surface.

4.6. Theme System Overview



Core tokens allow swapping palettes (industrial sepia → alternative themes) without structural CSS changes.

4.7. Deployment Workflow



Local authoring → version control → optional CI → static host (GitHub Pages/Netlify/Vercel) → CDN delivery.



Looking for all diagrams on one page? See [Architecture Diagrams Gallery](#).

4.8. Section Listing (CSS Layers)

The stylesheet is segmented for maintainability:

- CSS Variables (design tokens)

- Global Styles (reset + body)
- Header Styles
- Logo Styles
- Navigation Styles
- Content Area Styles
- Watermark Background Styles
- Image Gallery Styles
- Footer Styles

Each segment consumes tokens from `:root` for consistent theming.

5. Features in Detail

5.1. Responsive Design

The website adapts to different screen sizes using:

- Flexbox layouts
- Flexible images (`max-width: 100%`)
- Viewport meta tag
- Wrapped gallery for mobile devices

5.2. Accessibility

WCAG compliance features:

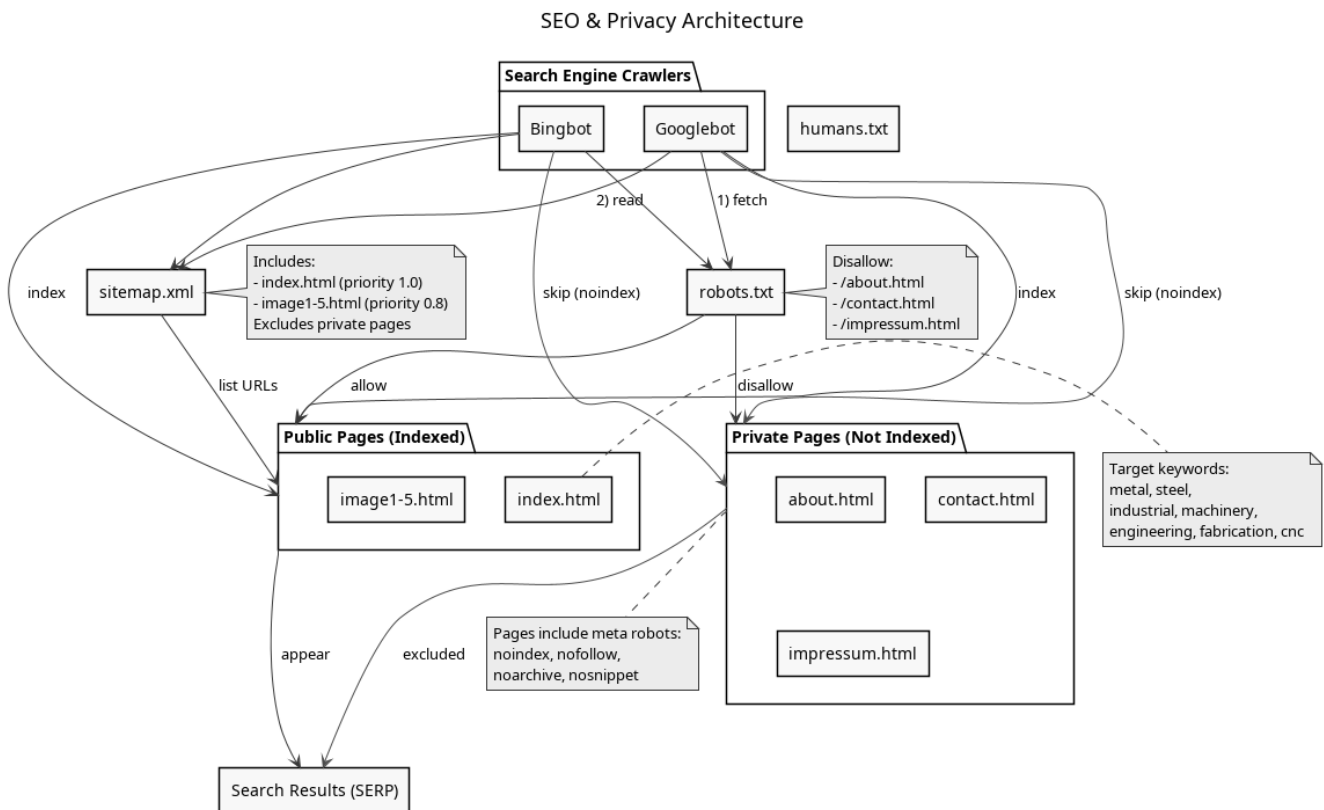
- Semantic HTML5 elements (`<header>`, `<nav>`, `<main>`, `<footer>`, `<article>`)
- ARIA labels for navigation and content sections
- Proper heading hierarchy (h1 → h2)
- Descriptive alt text for images
- Keyboard navigation support
- Sufficient color contrast ratios

5.3. SEO Optimization

Search engine optimization includes:

- Meta descriptions on all pages
- Keyword targeting: metal, steel, industrial, machinery, engineering, fabrication, cnc
- Semantic HTML structure

- **robots.txt** for crawler control
- **sitemap.xml** for search engines
- Proper page titles
- Author attribution



5.4. Privacy Compliance

Privacy-focused features:

- Three pages excluded from search engines (about, contact, impressum)
- **noindex, nofollow, noarchive, nosnippet** meta tags
- robots.txt disallow directives
- Legal notice page (impressum)

5.5. CSS Variables Theming

Easy theme customization through CSS variables:

```

:root {
  /* Colors */
  --color-primary: #3d2f21;
  --color-background: #f4f4f4;

  /* Spacing */
  --spacing-sm: 20px;

```

```
--spacing-md: 40px;

/* Logo */
--logo-size: 80px;

/* Gallery */
--image-size: 250px;
--image-rotation-large: 6deg;
}
```

See [Theme Variations Guide](#) for alternative color schemes.

6. Customization Guide

6.1. Changing Colors

Edit the CSS variables in `css/style.css`:

```
:root {
  --color-primary: #1a3a52;      /* New header/footer color */
  --color-background: #f0f4f8;  /* New background color */
}
```

6.2. Changing Logo

1. Replace `images/gear.png` with your logo (recommended: 80x80px)
2. Replace `images/gear.ico` with your favicon
3. Update the CSS variable if using different size:

```
:root {
  --logo-size: 100px; /* Adjust as needed */
}
```

6.3. Adding New Pages

1. Create new HTML file in `html/` directory
2. Copy structure from existing page
3. Update navigation links in all files
4. Add to `sitemap.xml` if publicly accessible
5. Update `robots.txt` if page should be excluded

6.4. Modifying Gallery

Change image count or arrangement in `index.html`:

```
<section class="image-gallery" aria-label="Project Gallery">
  <article class="image-container rotate-left">
    <a href="image1.html">
      
    </a>
  </article>
  <!-- Add more images -->
</section>
```

Rotation classes available:

- `rotate-left` (-6deg)
- `rotate-left-center` (-3deg)
- `rotate-center` (0deg)
- `rotate-right-center` (+3deg)
- `rotate-right` (+6deg)

7. Deployment

7.1. Static Hosting

Deploy to any static hosting service:

7.1.1. GitHub Pages

```
# Push html/ folder to gh-pages branch
git subtree push --prefix html origin gh-pages
```

7.1.2. Netlify

1. Connect your GitHub repository
2. Set build directory to `html`
3. Deploy

7.1.3. Vercel

1. Import your GitHub repository
2. Set output directory to `html`

7.2. Apache Configuration

If using Apache, add `.htaccess` for security headers:

```
# Security Headers
Header set X-Frame-Options "SAMEORIGIN"
Header set X-Content-Type-Options "nosniff"
Header set X-XSS-Protection "1; mode=block"
Header set Referrer-Policy "strict-origin-when-cross-origin"

# Enable compression
<IfModule mod_deflate.c>
    AddOutputFilterByType DEFLATE text/html text/css text/javascript
</IfModule>

# Browser caching
<IfModule mod_expires.c>
    ExpiresActive On
    ExpiresByType image/jpeg "access plus 1 year"
    ExpiresByType image/png "access plus 1 year"
    ExpiresByType text/css "access plus 1 month"
</IfModule>
```

7.3. Pre-Deployment Checklist

Before deploying to production:



1. ☐ Replace placeholder images with licensed images
2. ☐ Update impressum.html with actual company information
3. ☐ Change domain in sitemap.xml
4. ☐ Test all links and navigation
5. ☐ Validate HTML (<https://validator.w3.org/>)
6. ☐ Validate CSS (<https://jigsaw.w3.org/css-validator/>)
7. ☐ Test accessibility (<https://wave.webaim.org/>)
8. ☐ Test on multiple browsers and devices
9. ☐ Check mobile responsiveness
10. ☐ Verify robots.txt configuration

8. Maintenance

8.1. Regular Updates

- Update project images and descriptions
- Refresh meta descriptions for SEO
- Check for broken links
- Update copyright year in footer
- Review and update CHANGELOG.md

8.2. Browser Testing

Test regularly in:

- Google Chrome (latest)
- Mozilla Firefox (latest)
- Apple Safari (latest)
- Microsoft Edge (latest)
- Mobile browsers (iOS Safari, Chrome Mobile)

8.3. Performance Optimization

1. Optimize images (compress, use WebP)
2. Minify CSS for production
3. Enable server compression (gzip)
4. Implement browser caching
5. Use a CDN for static assets

9. Troubleshooting

9.1. Common Issues

9.1.1. Images Not Loading

Problem: Images don't display on live site

Solution: Check file paths are relative and case-sensitive

```
<!-- Correct -->  

```

```
<!-- Incorrect -->


```

9.1.2. Fixed Footer Overlapping Content

Problem: Footer covers bottom content

Solution: Content has `padding-bottom: 100px` to prevent overlap

9.1.3. Logo Position Wrong

Problem: Logo doesn't align properly

Solution: Adjust CSS variables:

```
:root {
  --logo-position-left: 20px; /* Adjust horizontal position */
  --logo-size: 80px;         /* Adjust size */
}
```

9.1.4. Gallery Not Responsive

Problem: Images overflow on mobile

Solution: Gallery uses `flex-wrap: wrap` by default. Check CSS variables:

```
:root {
  --image-size: 250px; /* Reduce for smaller screens */
  --gallery-gap: 40px; /* Reduce spacing */
}
```

10. Contributing

See [CONTRIBUTING.md](#) for contribution guidelines.

10.1. Code Style

- Use 4 spaces for indentation
- Add comments for complex sections
- Follow semantic HTML conventions
- Maintain accessibility features
- Test across browsers

11. License

This project is licensed under the MIT License. See [LICENSE](#) file for details.



Image Licensing Notice:

The placeholder images (metal1-5.jpg, gear.png, gear.ico) are NOT covered by the MIT License and must be replaced with properly licensed images before commercial use.

12. Support

- GitHub Issues: <https://github.com/iosan/webup/issues>
- Repository: <https://github.com/iosan/webup>
- Branch: v0.1.0

13. Appendix

13.1. File Reference

13.1.1. index.html

Homepage with 5-image project gallery

13.1.2. about.html

Company information page (excluded from search engines)

13.1.3. contact.html

Contact information page (excluded from search engines)

13.1.4. impressum.html

Legal notice required in Germany (excluded from search engines)

13.1.5. image1-5.html

Individual project detail pages with watermarked backgrounds

13.1.6. robots.txt

Directives for web crawlers

13.1.7. sitemap.xml

XML sitemap for search engines

13.1.8. humans.txt

Credits and technology information

13.2. Keywords

The website targets these SEO keywords:

- metal
- steel
- industrial
- machinery
- engineering
- fabrication
- cnc

13.3. Browser Support

Browser	Minimum Version
Chrome	Latest
Firefox	Latest
Safari	Latest
Edge	Latest
Opera	Latest

Last Updated: November 6, 2025

Version: 0.1.0

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